

Performance Testing Report — Week 4

DevelopersHub Corporation

Software Quality Assurance (SQA) Internship

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Application: DemoQA Web Application

1. Objective

The objective of this performance testing activity was to evaluate the performance, stability, and response behavior of the DemoQA Book Store API under load conditions using Apache JMeter. The test was conducted to analyze response time, throughput, and error handling when multiple virtual users access the system simultaneously.

2. Test Environment

Component	Details
Application	DemoQA Book Store API
URL Tested	https://demoqa.com/BookStore/v1/Books
Protocol	HTTPS
Request Method	GET
Tool Used	Apache JMeter
Operating System	Windows
Virtual Users	50 Users

3. Load Test Scenario

Parameter	Value
Number of Threads (Users)	50
Ramp-Up Period	10 Seconds
Loop Count	1
Request Type	GET
Endpoint Tested	/BookStore/v1/Books

4. Test Execution

The load test was executed successfully using Apache JMeter. A GET request was sent to the DemoQA Book Store API endpoint using 50 virtual users. Multiple requests were processed during the execution, and performance metrics were collected using JMeter listeners such as Summary Report, Graph Results, and View Results Tree.

5. Observations

- A total of 50 requests were executed successfully during the load test.
- The API handled all virtual user requests without any functional failure.
- The error percentage recorded during testing was 0.00%, which indicates successful request execution.
- The average response time observed during the test was approximately 6477 ms. (6.5 seconds).
- The minimum response time recorded was approximately 1078 ms. while the maximum response time reached approximately 30265 ms. under load conditions.
- Response times showed noticeable variation during execution, indicating that some requests required more processing time under load.
- The throughput recorded during testing was approximately 1.5 requests per second.
- Valid JSON response data containing book information was successfully returned by the API.
- The application remained stable and responsive during the execution of the performance test.

6. Graphs & Results

Summary Report Analysis

The Summary Report showed that all 50 requests were executed successfully with a 0% error rate. The average response time recorded was approximately 6477 ms. while the fastest request completed in approximately 1078 ms. the slowest request took approximately 30265 ms. during execution. The standard deviation value indicated variation in response times across different requests. Throughput was recorded at approximately 1.5 requests per second.

Graph Results Analysis

The Graph Results listener displayed variation in response times during the load test. Multiple response time spikes were observed while executing requests under the configured load of 50 virtual users. Some requests completed quickly, while others required higher response time under peak load conditions. Despite the variation, all requests were processed successfully without failures or errors.

View Results Tree Analysis

The View Results Tree confirmed that the GET request was successfully sent to the DemoQA Book Store API endpoint:

/BookStore/v1/Books

The listener displayed successful request execution with green status indicators. The API returned a valid 200 OK response along with JSON formatted response data containing book details such as title, author, ISBN, publisher, and pages. This confirmed that the endpoint functioned correctly during load execution.

7. Conclusion

The performance test was executed successfully using Apache JMeter with 50 virtual users. The application handled all requests without failures and maintained a 0% error rate throughout execution. Although response time variation was observed during load conditions, the API remained functional and responsive. Overall, the DemoQA Book Store API performed successfully under the configured load scenario.

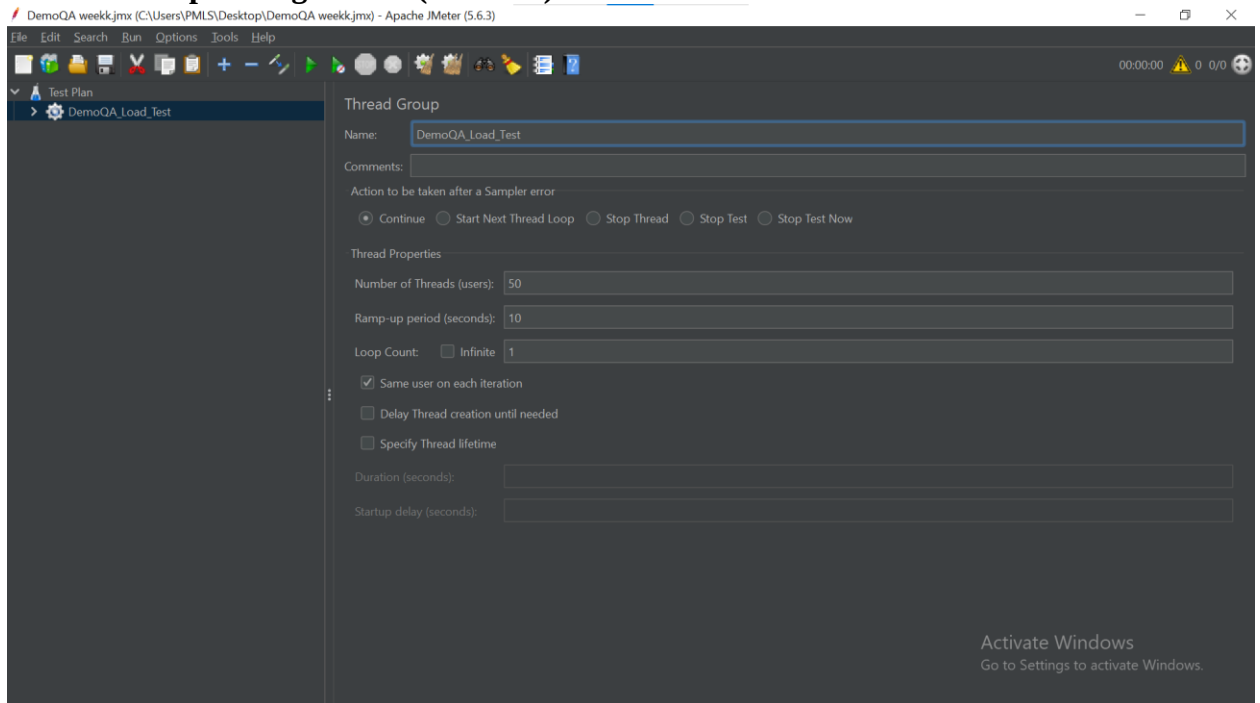
8. Recommendations

- Response time optimization may improve overall system performance.
- Additional testing with higher user loads can be performed for deeper performance analysis.
- Monitoring server-side performance may help identify delays during peak load conditions.

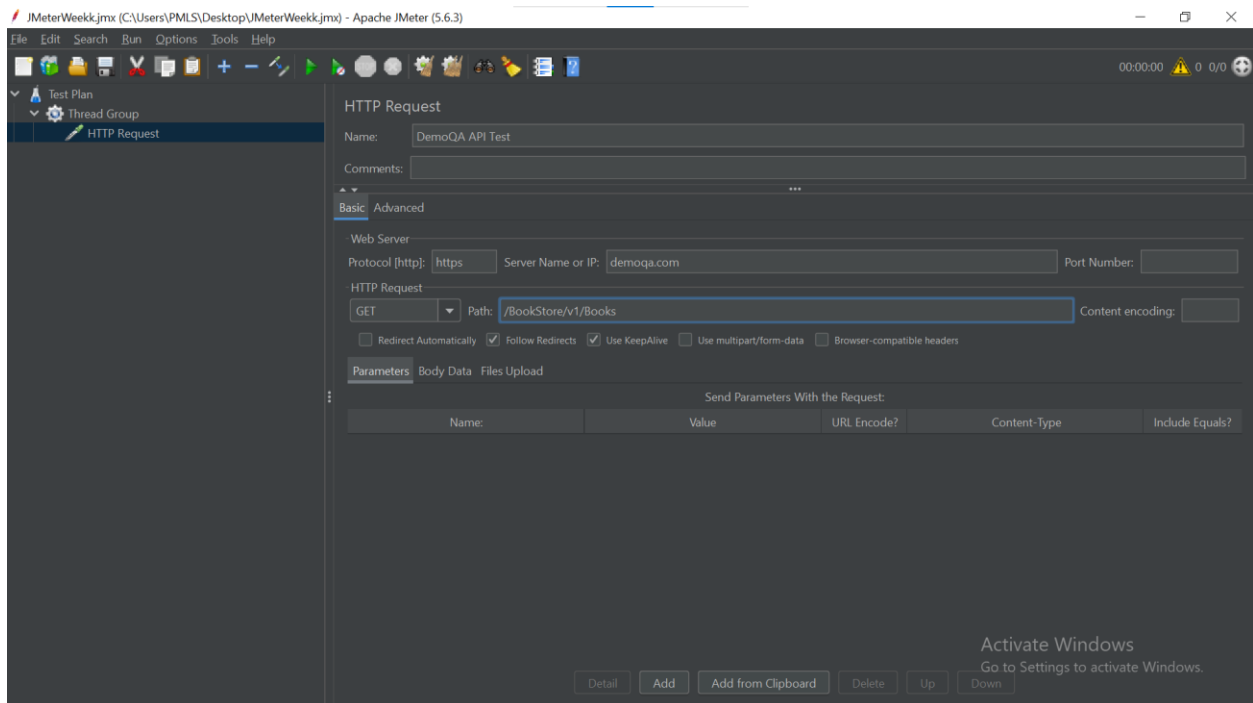
9. Attachments

The following screenshots are attached as evidence of test execution:

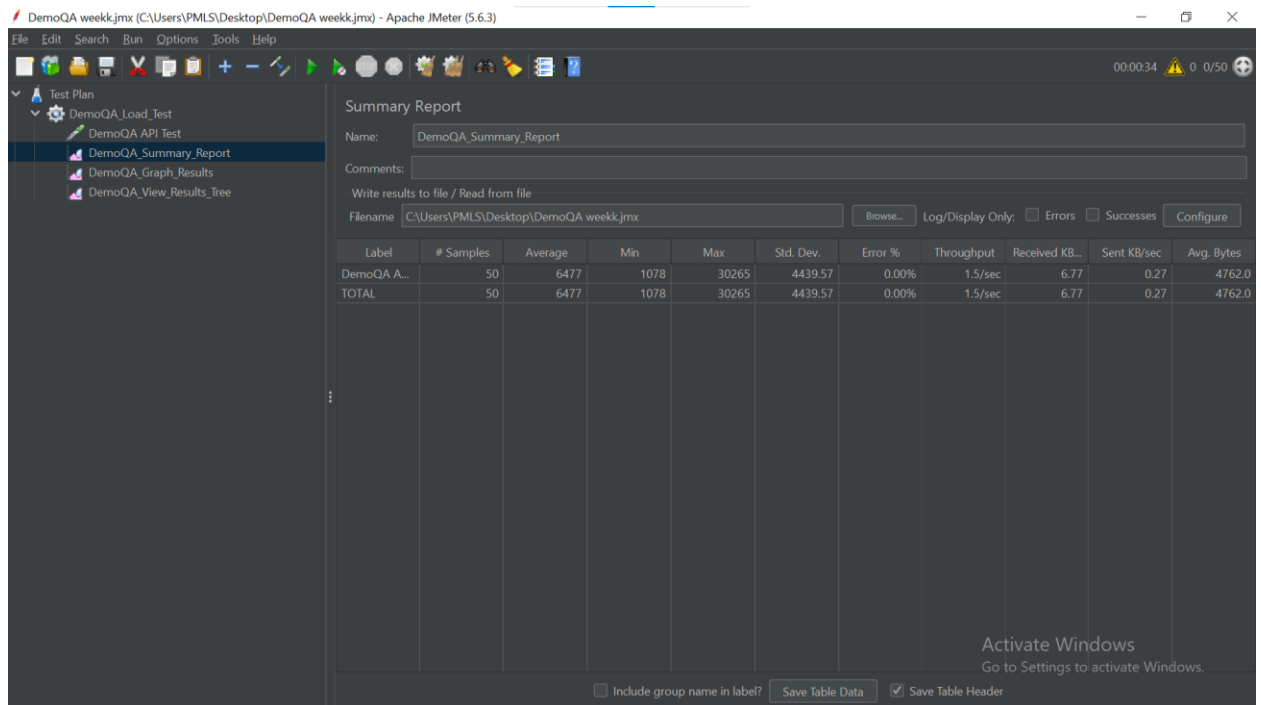
- **Thread Group Configuration (50 Users)**



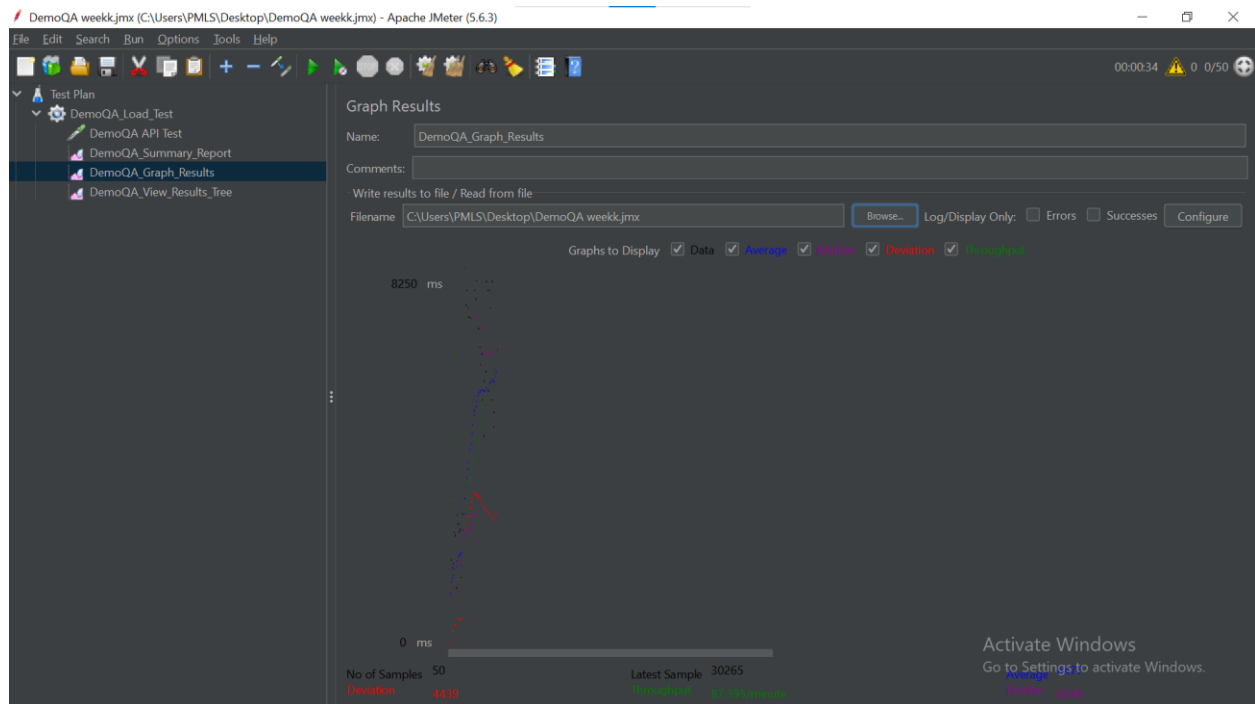
- **HTTP Request Setup**



- **Summary Report**

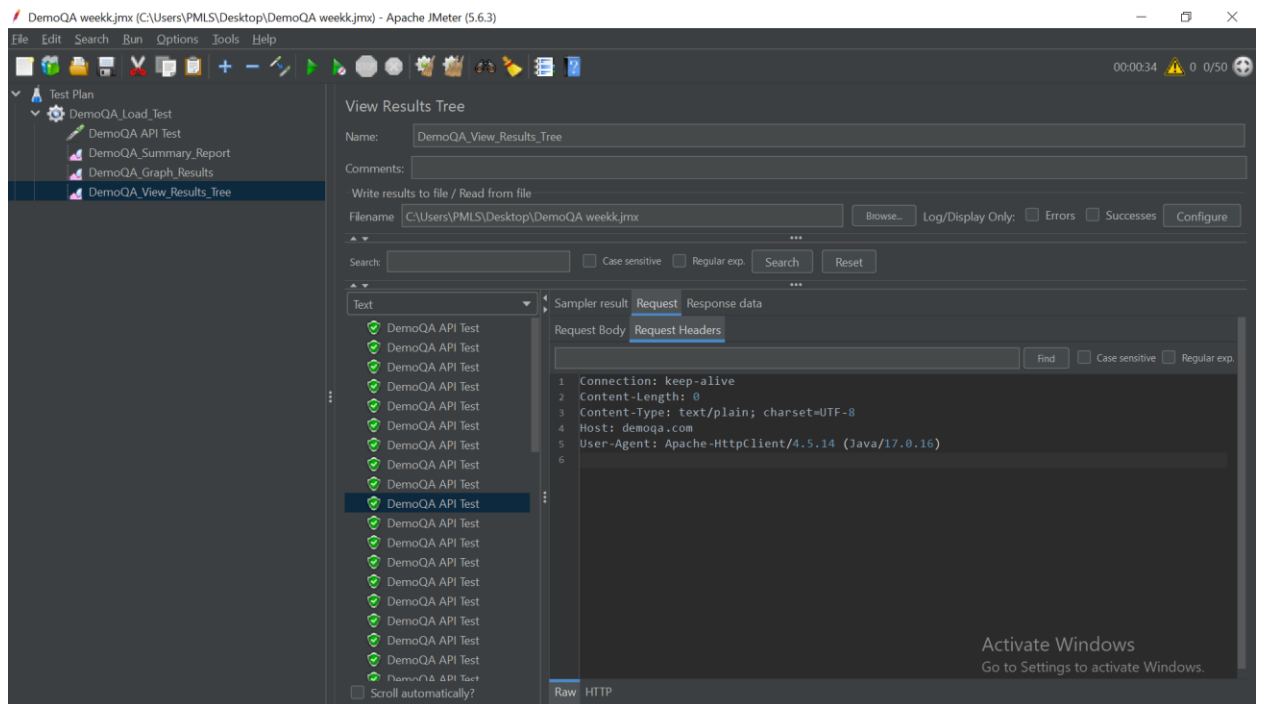


- **Graph Results**

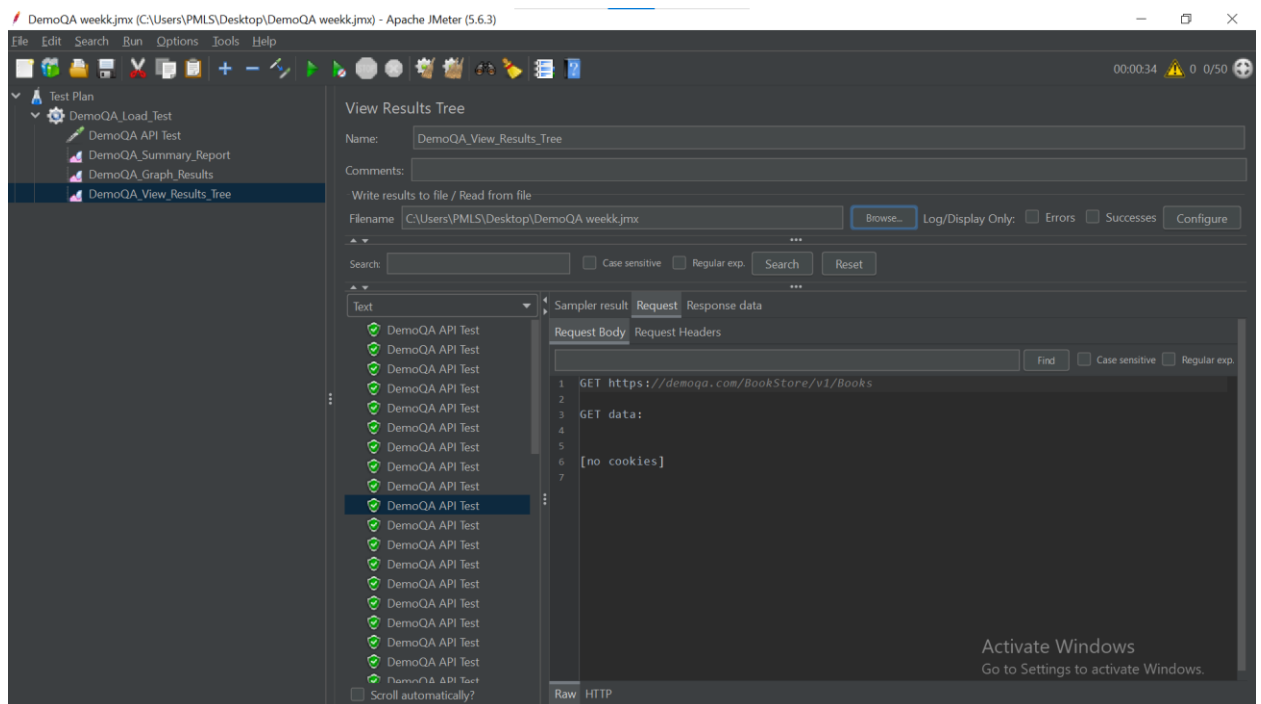


- **View Results Tree**

❖ Request Data(headers)



❖ Request Data(body)



❖ Response Data

The screenshot displays the Apache JMeter 5.6.3 application window. The title bar indicates the file 'DemoQA weekk.jmx' is open at 'C:\Users\PMLS\Desktop\DemoQA weekk.jmx'. The interface is divided into several sections:

- Test Plan:** A tree view on the left showing the test structure, including 'DemoQA Load Test', 'DemoQA API Test', 'DemoQA_Summary_Report', 'DemoQA_Graph_Results', and 'DemoQA_View_Results_Tree'.
- View Results Tree:** The main panel on the right, showing details for the selected 'DemoQA_View_Results_Tree'. It includes fields for Name, Comments, and File name (C:\Users\PMLS\Desktop\DemoQA weekk.jmx).
- Response data:** The 'Response data' tab is selected, showing the 'Response Body' as a JSON array of book objects. The JSON content is visible in the text area.

The JSON response body contains an array of book objects, each with fields like 'isbn', 'title', 'subTitle', 'author', 'publish_date', 'pages', and 'description'. The first object is for 'Git Pocket Guide' by Richard E. Silverman, and the last is for 'You Don't Know JS' by Kyle Simpson.